INCH-POUND

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PERFORMANCE SPECIFICATION

LEATHER, CATTLEHIDE, FOR FOOTWEAR UPPERS AND GUSSET, CHROME TANNED, FATLIQUORED

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers smooth finish, full grain and corrected grain, chrome tanned, fatliquored, cattlehide leather for footwear uppers and gusset.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FNSE, 2800 South 20th Street, Philadelphia, PA 19145-5099, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8330

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1.2 Classifiication. The leather shall be of two types in the following treatments as specified (see 6.2).

Type I - Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored

Type II - Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored (see 6.4)

Treatment A - Treated for resistance to water absorption
Treatment B - Treated for resistance to water penetration (Type I)
Treatment C - Not treated for resistance to water
Treatment A/B - Treated for resistance to water absorption and penetration (Type I)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing

(Unless otherwise indicated, copies of Federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association Of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 22709.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D2098 - Test Method for Dynamic Water Resistance of Shoe Upper Leather by the Dow Corning Leather Tester

D2099 - Test Method for Dynamic Water Resistance of Shoe Upper Leather by the Maeser Water Penetration Tester

D2212 - Test Method for Slit Tear Resistance of Leather

D2807 - Test Method for Chromic Oxide in Leather (Perchloric Acid Oxidation)

D2810 - Test Method for pH of Leather

D4705 - Test Method for Stitch Tear Strength of leather, Double Hole

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection, in accordance with 4.2.

3.2 Material. The leather shall be full grain or corrected grain, chrome tanned cattlehide sides.

3.3 Staining. Except as noted below, the flesh side of the leather shall be tested for staining in accordance with 4.4. The stain on the test pad shall have an expanded AATCC value of not less than 3.

Note: Leather which has a natural colored flesh side that has been subject to small amounts of dye added to the drum in order to level the shade of the natural color need not to be tested for straining.

3.4 Finish. Corrected grain leather shall be buffed or snuffed only to the extent that is necessary to remove superficial blemishes. The grain side of the leather shall have a light application of finish containing only sufficient pigment to assist in making the color uniform. The finish of the grain side shall be smooth.

Note: A fine hair cell embossed pattern may be applied to the grain surface of the side. Printed leather shall not be permitted. If embossed, the leather shall match the standard sample for fine hair cell pattern. Leather previously dyed or finished in a color not specified and then redyed or refinished to match a specified color shall not be acceptable.

3.5 Fungicide. Unless other wise specified (see 6.2), one of the following leather fungicides is required.

3.5.1 Paranitrophenol. The leather shall contain not less than 0.18 and not more than 0.70 percent paranitrophenol fungicide (see 4.5.1).

3.5.2 Vitasan. The leather shall not contain not less than 1.6 and not more than 5.0 percent Vitasan with the active ingredient ortho-phenylphenol.

3.5.3 Amical. The leather shall contain not less than 300 ppm and not more than 2000 ppm of active ingredient
Diiodomethyl-para-tolylsulfone (see 4.5.3).

3.6 Water resistant requirement. When treatment A,B or A/B is specified, the leather shall meet the following requirements as applicable.

3.6.1 Treatment A (water absorption) Types I & II. When treatment A is specified, specimens of the treated leather shall gain not more than 30 percent of their weight by water absorption when tested as specified in 4.4.

3.6.2 Treatment B (water penetration) Type I only. When treatment B is specified, at least 80 percent of the specimens tested shall withstand not less than 15,000 cycles by ASTM method

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D2099 or not less than 4,000 cycles by ASTM method D2098 without water penetration, when tested as specified in 4.4.

3.6.3 Treatment A/B (water absorption/penetration) Type I only. When treatment A/B is specified, paragraphs 3.6.1 and 3.6.2 shall apply.

3.7 Tearing strength (slots tear) Type I. At least 20 percent of the specimens tested can fall below the 28 pounds but none below 20 pounds.

3.8 Stitch tearing strength Type II. At least 80% of the specimens tested shall have stitch tearing strength values of not less than 30 pounds when tested as specified 4.4.

3.9 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally, preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.10 Chemical requirements. The leather shall conform to the chemical requirements in Table I when tested as specified in 4.4.

TABLE I. Chemical requirements

Characteristics	Minimum	Maximum
Chloroform soluble material, percent 1/	4.0	16.0
Chrome oxide shall be calculated on a hide substance basis, percent	4.0	7.5
Non-determined material, percent (Type I) 1/	-	20.0
pH value	3.0	4.0
Total ash, percent (Type I) 1/	-	10.0

1/ Calculated on a moisture-free basis.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.2)
2. Quality conformance inspection (see 4.3)

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4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be tested as specified in 4.4.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with the referenced documents.

4.3.1 Component and material inspection. The above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4 End item testing. The methods of testing specified in FED-STD-311, (and ASTM wherever applicable) and as listed in Table II shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311, (and ASTM wherever applicable). All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

a. More than three specimens fail requirements for water penetration or slot tearing strength.

b. Any specimen fails requirements for water absorption, or staining.

c. Any composite test result fails to meet the specified requirements.

TABLE II. End item tests

Characteristic	Requirement para.	Test method	Sample unit	Composition
Material identification	3.2		-	-
Tannage	3.2		-	-
Staining	3.3.	3021 1/ 4/	x	-
Fungicide	3.5	4.5	-	x
Water absorption (treatment A)	3.6.1	8112 4/	x	-
Water penetration (treatment B)	3.6.2	D2098 or D2099 2/ 5/	x	-
Tearing strength (slot tear)	3.7	D2212 5/	x	-
Stitch tearing strength	3.8	D4705 5/	x	-

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TABLE II. End item tests (continued)

Characteristic	Require-ment para.	Test method	Sample unit	Composi-tion
Chloroform-soluble material	3.10	6341 4/	-	x
Chromic oxide	3.10	D2807 5/	-	x
Non-determined material (Type I)	3.10	4.6.1	-	x
pH value	3.10	D2810 5/	-	x
Total ash (Type I)	3.10	6341 3/ 4/	-	x

1/ Except that the flesh side of the leather shall be in contact with the color transfer pad.

2/ The test method used shall be referenced in the test report.

3/ The procedure for determining water soluble material shall be omitted. The ash shall be determined on the chloroform extracted specimen.

4/ FED-STD-311

5/ Refers to ASTM standards.

4.5. Fungicide

4.5.1 Paranitrophenol. Testing to determine the concentration of paranitrophenol in the leather shall be conducted in accordance with Test Method 6711 of FED-STD-311.

4.5.2 Vitasan. Testing to determine ortho-phenylphenol concentration in leather shall be conducted in accordance with Test Method I, Type I of MIL-L-8067.

4.5.3 Amical. Testing and testing equipment (see 4.5.3.4) to determine the concentration of Amical in the leather shall be conducted using the High Pressure Liquid Chromatography (HPLC) test method as follows:

4.5.3.1 Grinding. Grind the leather in according to paragraph 6.3.7 of ASTM Method D2813 Standard Practice for Physical and Chemical Tests.

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4.5.3.2 Stock standard preparation. Weigh accurately approximately 100mg of Diiodomethyl-para-tolylsulfone Internal Reference Standard (IRS), into a 100ml volumetric flask. Add approximately 50ml of acetonitrile, stopper and sonicate to aid dissolution (about 1 minute). Dilute to volume with acetonitrile. Working standards preparation. Transfer 3ml aliquots of the stock standard 25, 50, 100, and 250ml volumetric flasks. Dilute to volume with acetonitrile. These standards contain approximately 12, 30, 60 and 120ppm of Diiodomethyl-para-tolylsulfone.

Note: A lower standard may be required for samples containing very low levels of Amical. Solutions of diiodomethyl-para-tolylsulfone must be stored in darkness to prevent degradation. Minimize exposure to light during preparation and use. The standard preparations may be retained one month.

4.5.3.3 Sample preparation. Accurately weigh approximately 3-5g of sample into a 150ml beaker. Add 75-100ml of acetonitrile. Cover samples with parafilm; mix 16 to 24 hours on a magnetic stir plate. The following day, allow samples to settle; remove as necessary. Note: Samples may have to be diluted further to fall within standard curve.

4.5.3.4 Testing equipment:

a. HPLC System with variable UV detector and integrating computer. Suitable syringe and glassware.

b. A micro-Bondapak C18 column (Waters Associates P/N 2734) or equivalent.

c. 0.45-micron polycarbonate filters (Nuclepore Corporation, Catalog No. 111107) or equivalent.

d. Calculator capable of doing linear regression.

e. Reagents: Acetonitrile (Chromatographic grade).

4.5.3.4.1 Typical mobile phase. Mix 550ml of distilled water and 450ml of acetonitrile. Filter and degas the solution. The concentration may be varied to meet system suitability requirements within the range 50% to 40% acentonitrile.

4.5.3.4.2 Typical chromatographic conditions:

Injection Volume	20 Microliters
Flow	2.5 ml/minute
Detector	235 mm

4.5.3.5 System suitability. Inject the highest working standard until 2 percent agreement is reached between two successive injections. Chromatograph for approximately 15 minutes. Determine that after the solvent front one impurity peak elutes before the major peak. Elution for monoiodomethyl-para tolylsulfone and diiodomethyl-para-tolylsulfone is about 8.5 minutes.

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4.5.3.6 Procedure. Inject each standard and sample preparation in duplicate allowing each injection to run for at least 15 minutes. Measure the peak response for the diiodomethyl-para-tolylsulfone.

Note: The run time may have to be increased for samples with late eluting peaks. Perform a linear regression of peak response (area) versus concentration for each of the components.

For each component:
Concentration(ppm) in leather, concentration (ppm) from liner regression.

$$X \quad \frac{75}{\text{Sample wt (g)}}$$

4.6 Methods of inspection

4.6.1 Non-determined material. The percent non-determined material shall be calculated as follows:

Non-determined material, percent = 100 - (A + B + C)

Where: A = Percent chloroform extractable material determined as specified in Table II.
B = Percent ash determined as specified in Table II.
C = Percent hide substance determined as specified Method 6441 of FED-STD-311.

5. PACKAGING

5.1 Packaging. Commercial packaging practices shall be used when shipping the finished leather hides from subcontractor to prime contractor.

6. NOTES

6.1 Intended use. The cattlehide leather will be used in the upper leather parts of general purpose footwear.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Treatment required (see 1.2).
c. DODISS. The specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When a first article is required (see 3.1, 4.2, and 6.3).
e. Color, shall be specified in end item footwear document.
f. Levels of preservation and packing (see 5.1).
g. When palletization is required (see 5.1).
h. When fungicides are not required (see 3.5).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Leather, cattlehide, gusset, chrome tanned, fatliquored Type II. The type II in this specification supersedes MIL-L-10867E, Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored (see 1.2).

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word) listing
Footwear
Leather

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA - CT

Project 8330-0194

Review activities:

Army - MD
Air Force - 82, 99

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6.1 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-3122H	960117

3. DOCUMENT TITLE

LEATHER, CATTLEHIDE, FOR FOOTWEAR UPPERS AND GUSSET, CHROME TANNED, FATLIQUORED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DSC-FNSE	(1) Commercial (215) 737-8079	(2) AUTOVON 444-8079
c. ADDRESS (Include Zip Code) PO BOX 60530 PHILADELPHIA, PA 19145-0530	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete.

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